

1 Solution — GIAM 6.2

Problem 1

1.1 Problem

Consider the relation S defined by

$$S = \{(x, y) : x \text{ is smarter than } y\}.$$

Is S symmetric or anti-symmetric? Explain.

1.2 Setup

Take S to be a relation on some population P of people, with “smarter than” read as a *strict* comparison — analogous to $<$ on the integers — so that “ x is smarter than y ” rules out “ y is smarter than x ” by ordinary usage. (The reading matters; see the remark at the end.)

Recall the two properties at issue, for a relation R on a set X :

- **Symmetric.** For all $x, y \in X$, $(x, y) \in R \implies (y, x) \in R$.
- **Antisymmetric.** For all $x, y \in X$, $[(x, y) \in R \text{ and } (y, x) \in R] \implies x = y$.

1.3 S Is Not Symmetric

Pick any two people $A, B \in P$ with A genuinely smarter than B . Then $(A, B) \in S$, but $(B, A) \notin S$: the strict reading forbids both directions holding at once. One counterexample defeats the universal statement, so S is **not symmetric**.

1.4 S Is Antisymmetric — Vacuously

Antisymmetry asks: whenever $(x, y) \in S$ and $(y, x) \in S$, must $x = y$?

Under the strict reading, the hypothesis is *never satisfied*: $(x, y) \in S$ already forces $(y, x) \notin S$, so no pair x, y has both directions in S . The conditional is vacuously true, and S is **antisymmetric**.

In fact, S satisfies the stronger property of **asymmetry**,

$$(x, y) \in S \implies (y, x) \notin S,$$

and asymmetric relations are automatically antisymmetric — the antisymmetric implication has an unsatisfiable hypothesis, exactly as above. So the sharper description is " S is asymmetric, hence antisymmetric, and not symmetric."

1.5 Remark — Reading of “Smarter Than”

The conclusion rests on reading “smarter than” as strict. Two natural alternatives, neither matching the textbook’s analogy to $<$:

- **Reflexive reading**, “at least as smart as.” Then $(x, x) \in S$ for every x ; symmetry can still fail, and antisymmetry becomes the substantive claim that mutual at-least-as-smart-ness forces equality — plausible only after identifying people who are equally smart, yielding a partial or total order on equivalence classes.
- **Equivalence reading**, “equally smart.” Then S is symmetric and *not* antisymmetric, since distinct people can be equally smart.

The natural reading is the strict one, and so the answer stands: S is **antisymmetric** (vacuously, in fact asymmetric) and **not symmetric**.